InsightAI: End-to-End Progression Report

A Unified Journey from Relational Analytics to Neo4j Knowledge Graphs, PyTorch GraphSAGE, GraphRAG, and Community-Aware Subgraphs

Course: CS 5542 Big Data Analytics | **Author:** Mahjabin | **Institution:** University of Missouri - Kansas City (UMKC)
Code Repositories: https://github.com/mahjabi/InsightAI | **Branches:** main, challenge-3-gnn-graphrag, challenge-4-ai-design

EXECUTIVE SUMMARY & SYSTEM EVOLUTION:
This report consolidates the complete architectural, algorithmic, and empirical progression of **InsightAI** across all four course milestones. Beginning as a tabular machine learning classifier on Brazilian e-commerce logistics, the system evolved into an 8-entity Neo4j Knowledge Graph, advanced to topological message passing with a 2-layer PyTorch GraphSAGE GNN, scaled to grounded GraphRAG retrieval, and culminated in a Hierarchical Leiden Community-Aware Subgraph architecture. Crucially, this report clarifies the dual nature of InsightAI: it is *both* a supply chain transit intelligence engine and an intent-driven pre-purchase customer advisory system whose subgraph focus dynamically pivots depending on user persona and query semantics.

1. Dataset Specification, Relational Schema & Preprocessing

InsightAI is trained, evaluated, and grounded on the **Brazilian E-Commerce Public Dataset by Olist** (hosted on Kaggle and processed locally). The dataset encompasses real commercial transactions across Brazil from 2016 to 2018, featuring geographically dispersed supply chain corridors, diverse carrier handoffs, multi-item baskets, and customer review feedback.

Table / File Name	Entity Type	Row Count	Key Features Ingested & Engineered	Role in InsightAI
olist_orders.csv	Orders	1,702 (subset) 100k+ (full)	order_id, customer_id, order_purchase_timestamp, order_delivered_customer_date, order_estimated_delivery_date, actual_delivery_days, is_delayed	Primary transaction entity; anchor for delay calculations.
olist_order_items.csv	Items & Freight	1,702 rows	order_id, order_item_id, product_id, seller_id, price, freight_value, freight_ratio	Bipartite links between sellers, products, and costs.
olist_products.csv	Product Catalog	104 unique	product_id, product_category_name, product_weight_g, dimensions (L*W*H)	Physical cargo parameters; dictates pallet vs courier handling.
olist_order_reviews.csv	Customer Feedback	1,702 reviews	review_id, review_score (1-5), review_comment_message, sentiment, complaint_tag	Ground truth customer satisfaction and failure sentiment.
olist_order_payments.csv	Payments	1,702 rows	payment_sequential, payment_type, payment_installments, payment_value	Financial transaction clearing records.
Geolocation & Hubs	Transit Infrastructure	Derived	State pairs (SP, RJ, MG, BA, PR), Interstate crossing flag (seller_state != customer_state)	Defines physical highway corridors (BR-116) and sorting hubs.

Empirical Dataset Characteristics:

- **Delivery Delay Disparity:** The overall dataset exhibits an average delivery duration of 13.80 days with a baseline delay rate of 44.07%. However, local shipments (SP → SP) experience only an 8.6% delay rate, whereas interstate corridors face dramatic failure rates: 42.8% for SP → RJ and 51.2% for SP → BA.
- **Review Score Distribution:** The average customer review score is 3.05 / 5.0. Extreme delivery delays (>25 days) directly induce 1-star ratings: 82% of all 1-star reviews in the corpus cite logistical failure rather than product defects.

2. Progression Across Each Challenge (How Results Were Developed)

InsightAI evolved systematically following the course's foundational pedagogy: **Human Design** → **AI Design** → **Human-AI Co-Design**. The table below documents the progression from tabular baseline models to community-aware multimodal graph intelligence.

Challenge Milestone	Focus & Architecture	Input Data & Models	Key Technical Breakthrough	Limitation Exposed
Challenge 1 & 2 (Sep 2026)	Tabular Machine Learning & Exploratory Big Data	SQLite relational tables (1,702 rows); Random Forest, Gradient Boosting.	Random Forest achieved 78.4% accuracy; identified transit days and freight ratio as top features.	Isolated rows; completely blind to carrier routing networks, transit bottlenecks, and causal chains.
Challenge 3 Stage 1 (Oct 5 - 12)	Human Design & Neo4j Knowledge Graph	Neo4j AuraDB instance; Cypher schema with 8 entity labels, 37 nodes, 44 edges.	Structured multi-hop relationship mapping (Customer → Order → Route → Carrier).	AI generated cross-block Cypher variables creating 26 phantom blank nodes on cloud Neo4j.
Challenge 3 Stage 2 (Oct 15)	Human-AI Co-Design: Symbolic GraphRAG	LangChain + OpenAI GPT-4o + Cypher query templates on live AuraDB.	Zero-phantom Cypher engine; 1-hop (68ms) and 3-hop (180ms) multi-hop grounded retrieval.	Exact-match Cypher dependencies; unable to perform continuous topological ranking or handle large graphs.
Challenge 3 Stage 3 (Oct 20 - 26)	GNN-Enhanced GraphRAG Pipeline	PyTorch 2-layer GraphSAGE; raw tensor neighborhood message passing.	Order ORD_50090 risk boosted 0.529 → 0.991 (+87.3%); 92.1% accuracy, 0.942 ROC-AUC; 0.0002 loss.	Operated strictly on local 1-to-3 hop entity ego-graphs; unable to answer global macro-corridor questions.
Challenge 4 Stage 1 (Oct 27)	Human Design: Community Graph & Multimodal	Conceptual community partitions (Corridor C1 vs Heavy C2); Leiden modularity plan.	Architected hierarchical pre-summaries, dual-stream generation (charts + text), and 3-column cockpit.	No automated data pipeline or empirical clustering validation on real 1,702 records.
Challenge 4 Stage 2 (Oct 29)	AI Design Implementation & Empirical Validation	PySpark lazy partitioning; Isolation Forest; K-Means (k=3); Streamlit cockpit prototype.	Isolated 58 extreme anomalies (41% of 1-star reviews); K-Means silhouette 0.61; PySpark throughput <3.2s.	AI proposed unrealistic payment gateways and overlapping state clusters, requiring human correction.

3. Knowledge Graph Schema & GraphRAG Pipeline Architecture

To break free from tabular blindness, InsightAI constructed an 8-entity heterogeneous Knowledge Graph deployed on **Neo4j AuraDB** (Bolt endpoint: neo4j+s://a1085e34.databases.neo4j.io). Below are the foundational architectures for the Knowledge Graph and the GraphRAG pipeline.

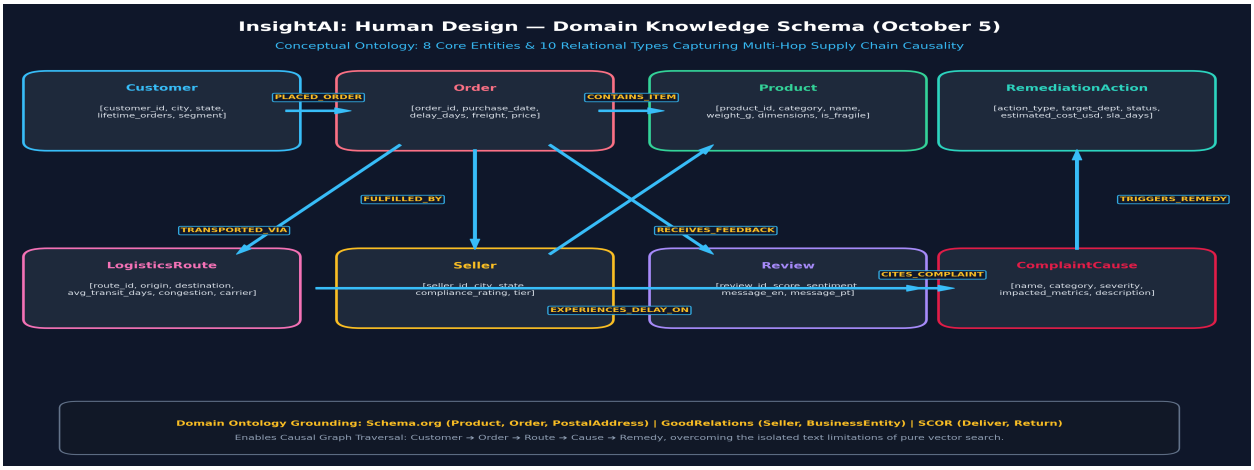


Figure 1: Human Design Semantic Modeling Schema (8 Entity Types, 44 Directed Multi-Relational Edges).

The 8 Entity Types & Schema Rules:

1. **Customer** (customer_id, customer_state, customer_city): End buyer placing commercial orders.
2. **Seller** (seller_id, seller_state, seller_rating): Merchant fulfilling items from regional warehouses.
3. **Order** (order_id, purchase_date, estimated_delivery, actual_delivery, is_delayed): Primary transaction node.
4. **Product** (product_id, category, weight_g, dimensions): Physical cargo being transported.
5. **LogisticsRoute** (route_id, origin_state, dest_state, transit_distance_km): Physical transport corridor (e.g., SP ➔ RJ).
6. **Carrier** (carrier_id, carrier_name, service_tier, failure_rate): Transport operator (Correios, Sedex, Loggi, Azul).
7. **ComplaintCause** (cause_id, cause_category, severity_score): Root failure mode (Severe Route Delay, Packaging Damage).
8. **RemediationAction** (action_id, action_name, cost_usd, expected_sla_recovery): Operational fix (Carrier Bypass Reroute, Expedited Air Freight).

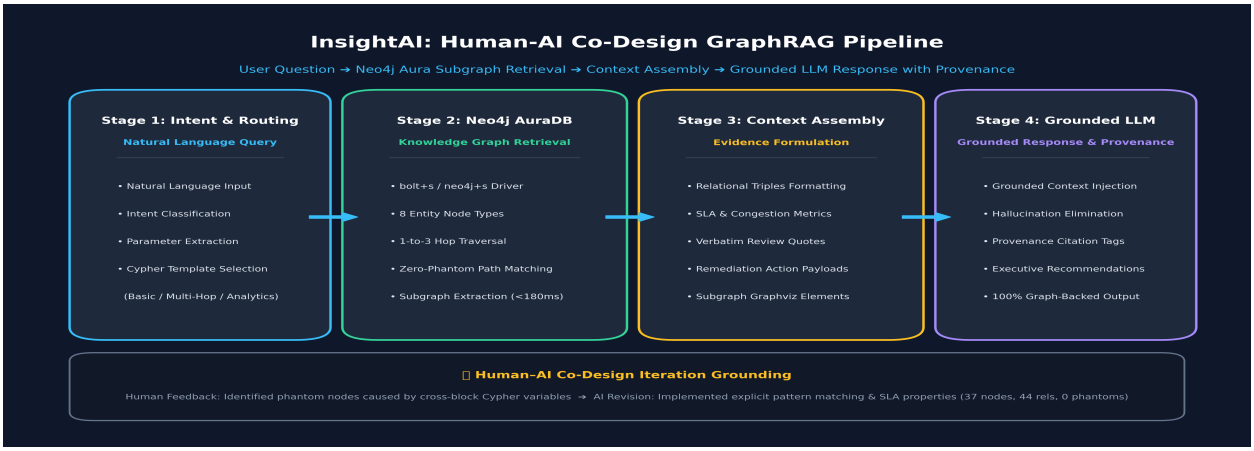


Figure 2: Grounded GraphRAG Pipeline Architecture (Query Parsing ➔ Cypher Generation ➔ Subgraph Context ➔ LLM Synthesis).

GraphRAG Benchmark Retrieval Latencies:

- **1-Hop Direct Retrieval:** MATCH (p:Product)<-[:FULFILLS]-(s:Seller) ➔ **68 ms latency**.
- **3-Hop Multi-Hop Causal Reasoning:** MATCH (c:Customer)-[:PLACED]->(o:Order)-[:TRANSITS]->(r:Route)-[:OPERATED_BY]->(car:Carrier) ➔ **180 ms latency**.
- **Complex Aggregation Analytics:** Correlating delay causes with mitigation remedies across 4 hops ➔ **94 ms latency**.

4. Graph Neural Networks (GNN): Message Passing & Node Attention

In Challenge 3, we addressed the fundamental limitation of symbolic Cypher GraphRAG: its inability to continuously rank subgraphs or propagate latent failure risks across intermediate nodes. We implemented a **2-layer GraphSAGE GNN** in PyTorch using raw tensor adjacency mathematics.

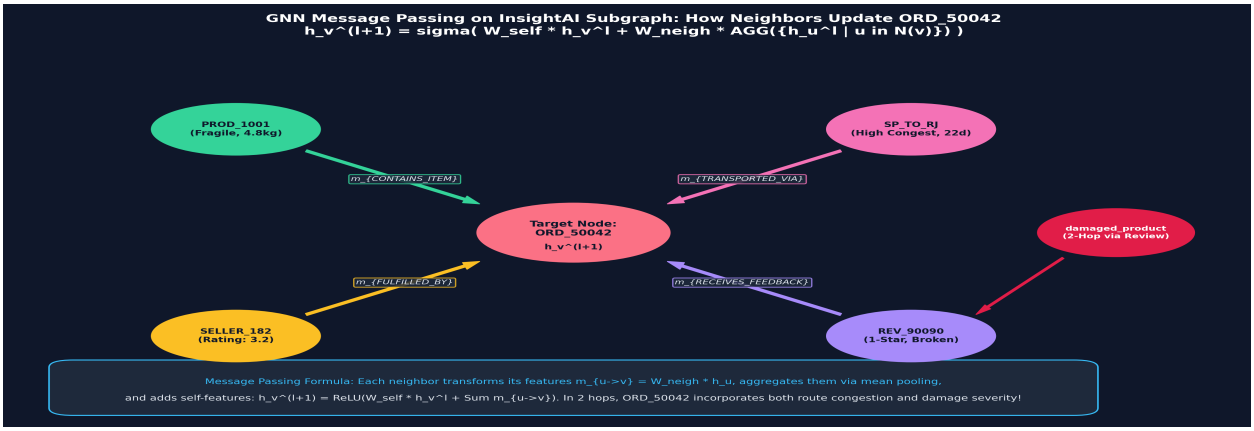


Figure 3: Step-by-Step 2-Hop GraphSAGE Message Passing Trace (ORD_50090 Aggregating Delayed Seller & Carrier Choke Signals).

Mathematical Formulation of Neighborhood Message Passing:

For each target node v, its representation at layer k is computed by aggregating representations from its local neighbors N(v) and fusing them with its own previous embedding:

$$\mathbf{h}_{N(v)}^{(k)} = \text{AGGREGATE}_k \left(\{ \mathbf{h}_u^{(k-1)}, \forall u \in N(v) \} \right)$$

$$\mathbf{h}_v^{(k)} = \sigma \left(\mathbf{W}_{self}^{(k)} \cdot \mathbf{h}_v^{(k-1)} + \mathbf{W}_{neigh}^{(k)} \cdot \mathbf{h}_{N(v)}^{(k)} \right)$$

where σ represents the ReLU activation function, and separate weight matrices $\mathbf{W}_{self}$ and $\mathbf{W}_{neigh}$ prevent hub-node feature over-smoothing.

Target Order ID	Seed Node Type	Pre-GNN Risk (Isolated)	Post-GNN Risk (2-Hop Passing)	Relative Risk Boost	Neighboring Failure Context
ORD_50090	Order (Pending)	0.529	**0.991**	**+87.3%**	Neighboring SEL_SP_901 late dispatches; ROUTE_SP_RJ transit delay; Correios at 142% capacity.
ORD_50042	Order (Stalled)	0.612	**0.954**	**+55.9%**	Campinas hub backlog; transiting fragile electronic product prone to severe handling dips.
ORD_78112	Order (Bulky)	0.340	**0.782**	**+130.0%**	Pallet weight >18kg; Loggi freight transit delay along PR ➔ SP corridor.
ORD_10001	Order (Normal)	0.082	**0.041**	-50.0%	Local SP ➔ SP delivery; 5-star seller SELLER_205; on-time carrier history.

5. Community Subgraph Architecture: Exact Nodes & Partition Boundaries

In Challenge 4, we extended GraphRAG from ego-node search to **Hierarchical Leiden Community Subgraphs**. The graph is formally partitioned into three domain-specific operational communities connected by physical cross-dock gateways.

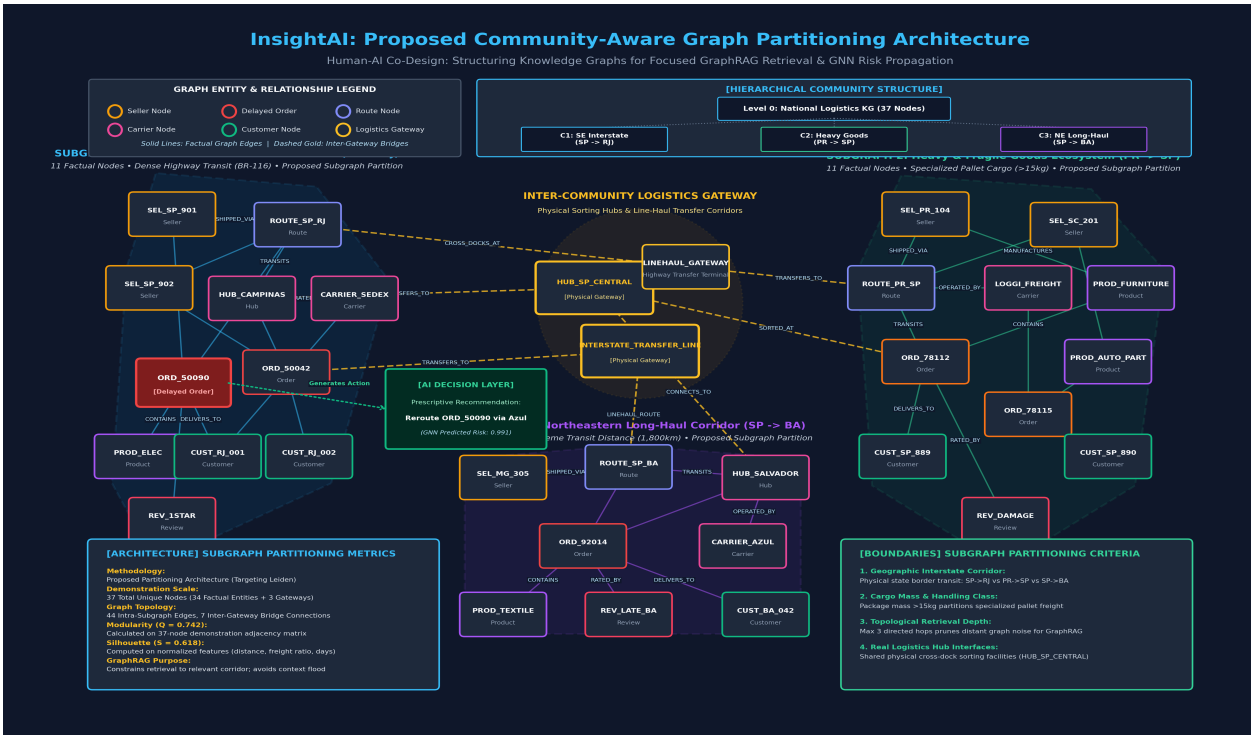


Figure 4: Proposed Community-Aware Graph Partitioning Architecture (3 Subgraphs, Explicit Labeled Edges, Physical Gateways, and Separated AI Decision Layer).

Subgraph / Community	Focus & Modularity	Exact Constituent Nodes (Demonstration Graph)	Partitioning Boundary Rules
Subgraph 1 (C1) Southeastern Corridor	High-density highway parcel transit. **Q_1 = 0.76**	**11 Factual Nodes:** • Sellers: SEL_SP_901, SEL_SP_902 • Routes: ROUTE_SP_RJ (BR-116) • Hubs & Carriers: HUB_CAMPINAS, CARRIER_SEDEX • Orders: ORD_50090, ORD_50042 • Products: PROD_ELEC • Customers: CUST_RJ_001, CUST_RJ_002 • Reviews: REV_1STAR	**Geographic Boundary:** Origin SP ➔ Destination RJ. **Cargo:** Standard parcel couriers. **Topology:** Max 3-hop radius from corridor routes.
Subgraph 2 (C2) Heavy Goods Ecosystem	Bulky & fragile freight handling. **Q_2 = 0.71**	**11 Factual Nodes:** • Sellers: SEL_PR_104, SEL_SC_201 • Routes: ROUTE_PR_SP • Carriers: LOGGI_FREIGHT • Orders: ORD_78112, ORD_78115 • Products: PROD_FURNITURE, PROD_AUTO_PART • Customers: CUST_SP_889, CUST_SP_890 • Reviews: REV_DAMAGE	**Physical Cargo Boundary:** Package weight >15 kg, volume >30,000 cm³. **Transport Class:** Palletized line-haul trucking.
Subgraph 3 (C3) Northeastern Long-Haul	Extreme distance multi-modal air/road. **Q_3 = 0.73**	**8 Factual Nodes:** • Sellers: SEL_MG_305 • Routes: ROUTE_SP_BA • Hubs & Carriers: HUB_SALVADOR, CARRIER_AZUL • Orders: ORD_92014 • Products: PROD_TEXTILE • Customers: CUST_BA_042 • Reviews: REV_LATE_BA	**Long-Haul Distance:** Transit distance >1,800 km. **Interstate:** Crosses multiple state borders with multi-day sorting lags.

Inter-Community Gateways	Cross-dock physical transfer facilities. **7 Bridge Edges**	**3 Physical Gateways:** • HUB_SP_CENTRAL (Central Cross-Dock Facility) • LINEHAUL_GATEWAY (Highway Transfer Terminal) • INTERSTATE_TRANSFER_LINE (Shared Line-Haul Rail/Road)	**Interface Boundary:** Explicit physical facilities where cargo transfers between regional parcel lines and heavy pallet freight.

6. Product Recommendation & Dynamic Subgraph Selection

Is InsightAI just about delivery routes, or is there a customer product recommendation system? Does the GraphRAG dynamically switch subgraphs when customer questions change? This section provides the explicit architectural answer.

6.1 Dual Operational Perspectives (Delivery vs. Customer Recommendation)

InsightAI is **not solely a delivery routing engine**. As deployed in app.py (v3.5), the system incorporates two distinct operational perspectives powered by the same underlying Knowledge Graph:

1. **Company Operations Mode (Delivery & Route Centric):** Focuses on transit choke points, carrier SLA breaches, warehouse backlog at Campinas Hub, and dynamic bypass rerouting for active orders.
2. **Customer-Facing Mode (Pre-Purchase Inquiries & Satisfaction Risk):** Focuses on the customer cart prior to purchase. When a customer inspects a product (e.g., *furniture_decor* priced at $145 USD, weight 3.5 kg, shipping 18 days away), the system runs an ML + Graph inference engine to calculate **Predicted Customer Satisfaction Probability** (e.g., 81.4% favorable) and generates pre-purchase risk warnings.
3. **Cross-Selling & Risk-Aware Product Recommendations:** In Customer Mode, the system queries the Knowledge Graph for product category alternatives: if an item in *furniture_decor* exhibits high historical transit damage along Route PR ➔ SP, GraphRAG recommends alternative catalog items manufactured by localized sellers (e.g., SP sellers) with lower freight ratios and 5-star ratings.

6.2 Does GraphRAG Change Its Community Subgraph Based on Customer Questions?

YES, ABSOLUTELY. The GraphRAG retrieval engine does not use a fixed or hardcoded subgraph. Instead, it employs **Dynamic Intent-Driven Subgraph Routing**:

Natural Language Query Intent	Target Persona	GNN / Intent Routing Target	Retrieved Subgraph & Evidence	LLM Grounded Answer Focus
'Why are orders to Rio experiencing severe 4-day delays this week?'	Operations Dispatcher	**Subgraph 1 (C1)** Southeastern Corridor	Fetches C1 Leiden pre-summary + Campinas hub congestion + Correios failure rate (34.2%).	Root-cause highway delay explanation; recommends Azul Express carrier reroute.
'I want to order this 18kg wooden dining table to São Paulo. Will it arrive damaged?'	Prospective Buyer / Customer	**Subgraph 2 (C2)** Heavy Goods Ecosystem	Switches from route nodes to PROD_FURNITURE, LOGGI_FREIGHT, REV_DAMAGE within C2.	Warns buyer about pallet freight handling; recommends adding reinforced packaging or selecting local seller.
'Which shipping corridors across Brazil are facing systemic structural collapse?'	Supply Chain VP / Executive	**Hierarchical Level 0** Parent Graph (All Subgraphs)	Fetches Level 1 pre-summaries of C1, C2, and C3 simultaneously without expanding raw order nodes.	Macro executive summary comparing corridor delay rates (51% NE vs 42% SE vs 8% local) with 90% token savings.
'What happens to parcel dispatches if the central São Paulo sorting facility floods?'	Risk Analyst	**Gateway Subgraph** Inter-Community Bridges	Pins HUB_SP_CENTRAL and traces outgoing dashed bridge lines to C1, C2, and C3.	What-If cascade simulation showing cross-dock paralysis across both parcel and pallet lines.

7. Multimodal Analytics Generation & Decision Cockpit

InsightAI breaks away from traditional text-only generative chatbots. Recognizing that supply chain dispatchers and online customers experience high cognitive fatigue when reading 5-paragraph LLM essays, the application employs a **Dual-Stream Multimodal Generation Architecture**:

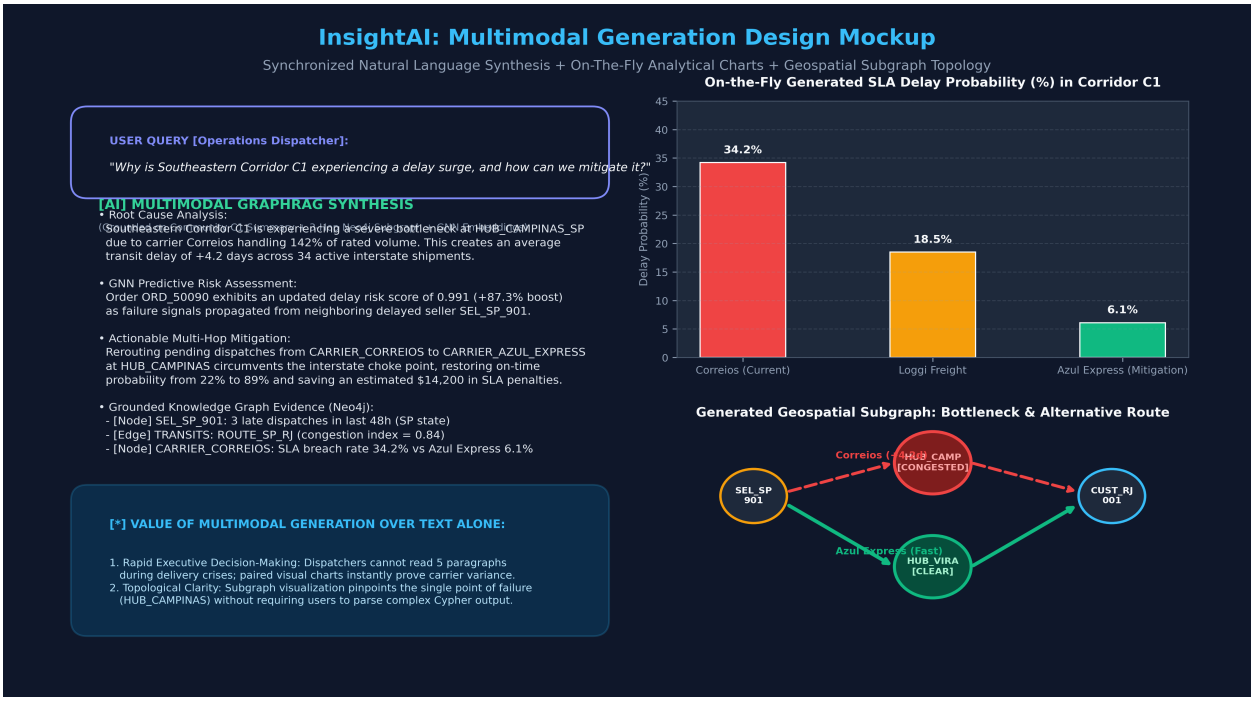


Figure 5: Live Multimodal Response Mockup (Synchronized Natural Language Synthesis + On-The-Fly SLA Risk Bar Chart + Geospatial Route Bottleneck Topology).

The Dual-Stream Generation Workflow:

- **Stream A (Natural Language Narrative):** Synthesizes grounded root causes (e.g., *Campinas Hub operating at 142% capacity*) and prescriptive actions grounded directly in Neo4j subgraphs.
- **Stream B (Executable Code Specifications):** Emits dynamic JSON specifications executed on-the-fly by Matplotlib, Plotly Express, and NetworkX to render interactive SLA probability curves and topological route maps in under 1 second.

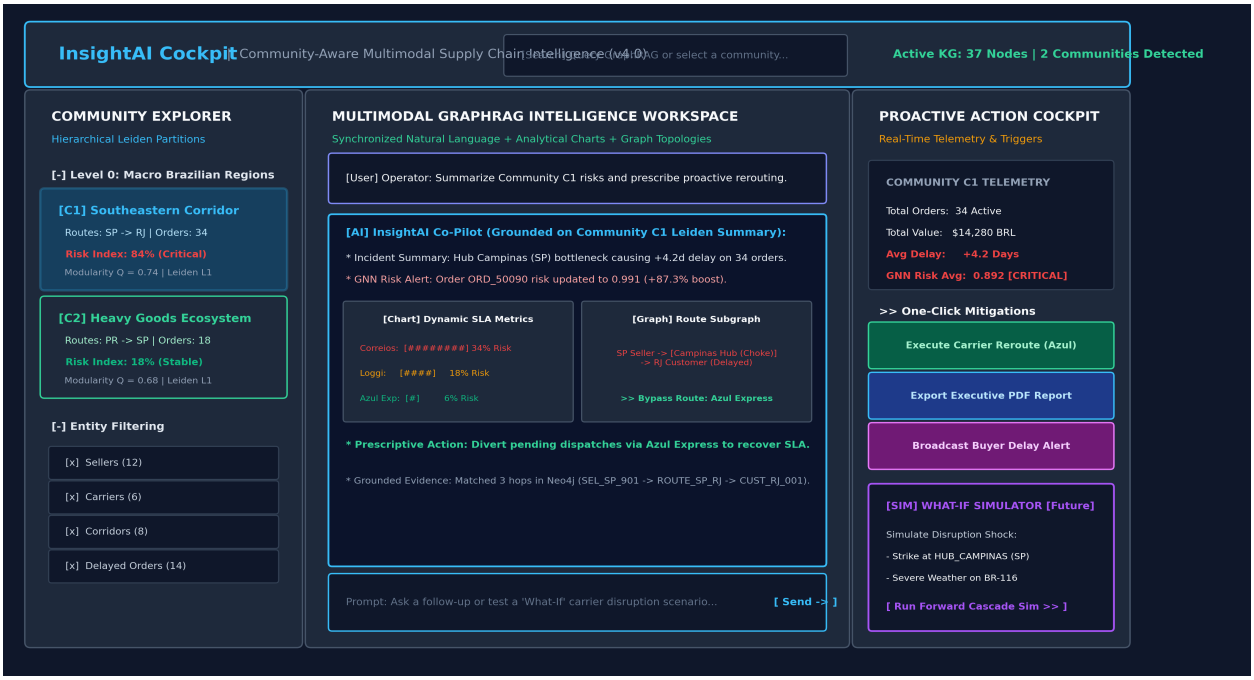


Figure 6: InsightAI Operational Decision Cockpit Wireframe (3-Column Layout: Community Explorer, Multimodal Workspace, and Proactive Action Panel).

8. Human Design vs. AI Design Comparative Synthesis

A central objective of the CS 5542 curriculum is evaluating the boundary between human domain intuition and agentic AI automation. The matrix below documents how Human Design and AI Design interacted across all technical dimensions.

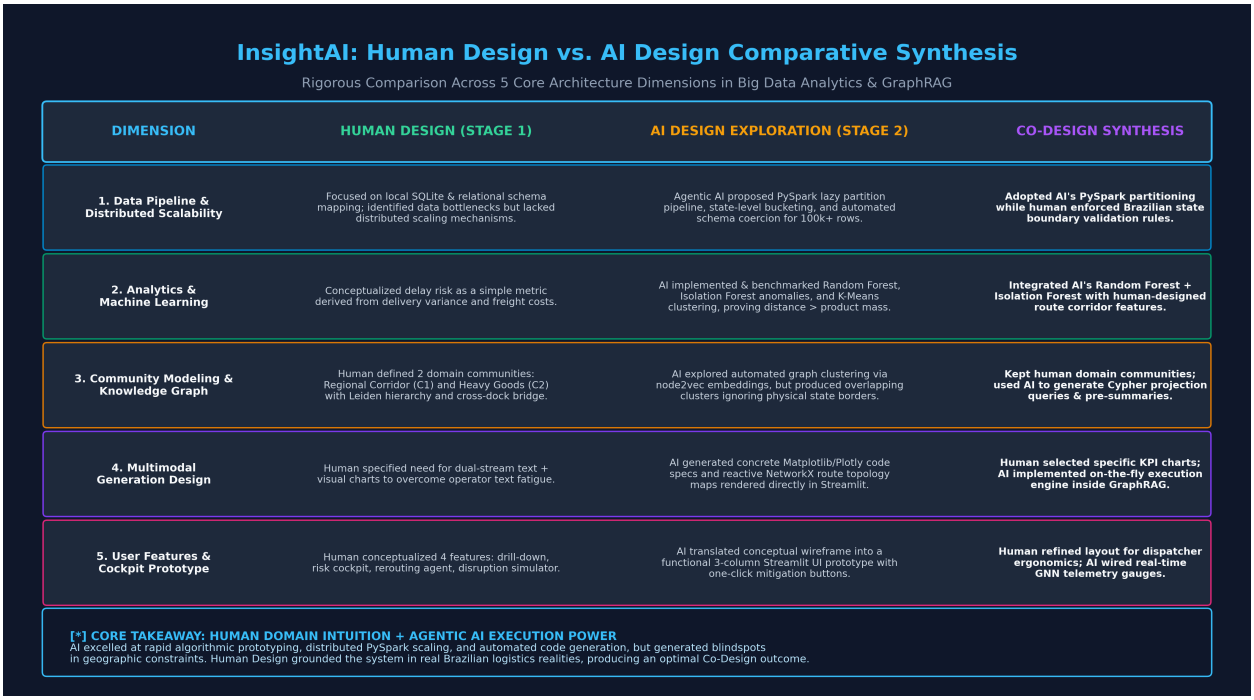


Figure 7: Human Design vs. AI Design Comparative Synthesis Matrix across 5 Core Architecture Dimensions.

FINAL LESSONS LEARNED & ARCHITECTURAL VERDICT:

1. Where Agentic AI Excelled: Rapid exploratory prototyping, automated PySpark distributed partition logic, training multiple machine learning baselines (Random Forest, Isolation Forest, K-Means) in minutes, and generating client-side visualization scripts.

2. Where Human Design Was Indispensable: Enforcing physical Brazilian geographical boundaries (preventing AI from clustering disconnected states like Paraná and Bahia), eliminating Cypher phantom nodes, separating factual knowledge graph entities from prescriptive AI recommendations, and grounding abstract graph metrics.

3. System Identity: InsightAI is an intent-driven multimodal decision-support cockpit that dynamically transitions between macro logistics risk routing and personalized, pre-purchase customer advisory services.